

31 Definition of definite Riemann integral

Definition 31.1. (partition)

Let $[a, b]$ be a closed interval where $a, b \in \mathbb{R}$.

A **partition** of $[a, b]$ is a finite subset of $[a, b]$ which contains a and b .

Let $P = \{a = x_0 < x_1 < \dots < x_{n-1} < x_n = b\}$ be a partition of $[a, b]$, where $n \in \mathbb{N}$.

Another partition Q is called a **refinement** of P if $Q \supset P$.

The **norm** of the partition P is defined to be $\|P\| = \max\{x_i - x_{i-1} : i = 1, \dots, n\}$.

Definition 31.2. (Riemann sum)

Let $f : [a, b] \rightarrow \mathbb{R}$ be a bounded function.

Let $P = \{a = x_0 < x_1 < \dots < x_{n-1} < x_n = b\}$ be a partition of $[a, b]$.

For each $1 \leq i \leq n$, let $M_i = \sup\{f(x) : x_{i-1} \leq x \leq x_i\}$, $m_i = \inf\{f(x) : x_{i-1} \leq x \leq x_i\}$,

and $t_i \in [x_{i-1}, x_i]$ and call t_i a **sample point**.

The number $U_f(P) = \sum_{i=1}^n M_i (x_i - x_{i-1})$ is called the **upper Riemann sum** of f about P .

The number $L_f(P) = \sum_{i=1}^n m_i (x_i - x_{i-1})$ is called the **lower Riemann sum** of f about P .

The number $R_f(P) = \sum_{i=1}^n f(t_i) (x_i - x_{i-1})$ is called a **Riemann sum** of f about P .

Proposition 31.3. (partitions and Riemann sums)

Let $f : [a, b] \rightarrow \mathbb{R}$ be a bounded function and let P and Q be partitions of $[a, b]$. Then

1. All of $L_f(P)$, $R_f(P)$ and $U_f(P)$ are well defined and $L_f(P) \leq R_f(P) \leq U_f(P)$.
2. If Q is a refinement of P , then $L_f(P) \leq L_f(Q)$ and $U_f(Q) \leq U_f(P)$.
3. $L_f(P) \leq U_f(Q)$.

Definition 31.4. (definite Riemann integral)

Let $f : [a, b] \rightarrow \mathbb{R}$ be a bounded function.

The two numbers

$$\sup_{P \text{ is a partition of } [a, b]} L_f(P) := \int_a^b f(x) dx \leq \int_a^b f(x) dx := \inf_{P \text{ is a partition of } [a, b]} U_f(P)$$

are called the **lower integral** and **upper integral** of f on $[a, b]$.

We say that f is **Riemann integrable** on $[a, b]$ if $\int_a^b f(x) dx = \int_a^b f(x) dx$;

in this case, the common value is called the **definite Riemann integral** of f on $[a, b]$

and is denoted by $\int_a^b f(x) dx$,

where the symbol $\int$ is an **integral sign**, the function f is called an **integrand**,

the numbers a and b are called **limits of integrations**,

while a is called the **lower limit** and b is called the **upper limit**,

and the letter x is a **dummy variable**.

We also use the usual convention that $\int_b^a f(x) dx = -\int_a^b f(x) dx$.

Example 31.5 (advanced exercise).

In the class of all functions that is positive on the rationals and zero on the irrationals in $[0, 1]$,

1. the characteristic function of $\mathbb{Q}$, is defined to be one on the rationals and zero on the irrationals is NOT Riemann integrable, and is nowhere continuous;
2. it can be modified by setting $f(p/q) = 1/q$ on rationals, also called as the Thomae function, to become Riemann integrable, is continuous on irrationals and discontinuous on rationals, and is nowhere differentiable;
3. it can further be modified to be differentiable on a countable subset of the irrationals;
4. it can further be modified to be differentiable on the non-Liouville irrationals and is not differentiable on the rationals and the set of the Liouville numbers which is uncountable dense and has measure zero.
5. it is not differentiable on an uncountable dense subset of the irrationals.

32 Properties on Riemann integrals and the Riemann-Lebesgue theorem

Theorem 32.1. (equivalent condition for Riemann integrability)

Let $f : [a, b] \rightarrow \mathbb{R}$ be bounded.

Then f is Riemann integrable on $[a, b]$

$\iff$ for any $\epsilon > 0$, there exists a partition P of $[a, b]$ such that $U_f(P) - L_f(P) < \epsilon$.

Theorem 32.2. (basic theorems for Riemann integrals)

Let $f, g : [a, b] \rightarrow \mathbb{R}$ be Riemann integrable on $[a, b]$. Then

1. (**linearity of Riemann integrals**)

$f \pm g$ and cf , where $c \in \mathbb{R}$ is a constant, are Riemann integrable on $[a, b]$,

and these Riemann integrals are given by

$$\int_a^b f(x) \pm g(x) dx = \int_a^b f(x) dx \pm \int_a^b g(x) dx \text{ and } \int_a^b cf(x) dx = c \int_a^b f(x) dx.$$

2. (**interval additive property**)

(a) If f is also Riemann integrable on $[b, c]$,

then f is Riemann integrable on $[a, c]$ and $\int_a^c f(x) dx = \int_a^b f(x) dx + \int_b^c f(x) dx$.

(b) If $[c, d] \subseteq [a, b]$, then f is Riemann integrable on $[c, d]$.

(c) If $c, d, e \in [a, b]$, then $\int_c^e f(x) dx = \int_c^d f(x) dx + \int_d^e f(x) dx$, regardless of the order of c, d, e .

3. (**comparison property**)

If $f(x) \leq g(x)$ for all $x \in [a, b]$, then $\int_a^b f(x) dx \leq \int_a^b g(x) dx$.

4. $|f|$ is Riemann integrable and $|\int_a^b f(x) dx| \leq \int_a^b |f(x)| dx$.

Theorem 32.3. (boundedness and monotonicity $\Rightarrow$ integrability)

If $f : [a, b] \rightarrow \mathbb{R}$ is monotone on $[a, b]$, then f is Riemann integrable on $[a, b]$.

Theorem 32.4. (continuity on a closed interval implies integrability)

If $f : [a, b] \rightarrow \mathbb{R}$ is continuous on $[a, b]$, then f is Riemann integrable on $[a, b]$.

Theorem 32.5 (the integrability theorem).

If $f : [a, b] \rightarrow \mathbb{R}$ is bounded on $[a, b]$ and is continuous at all except finitely many points in $[a, b]$, then f is Riemann integrable on $[a, b]$.

Definition 32.6. (content zero)

A set $Z \subseteq \mathbb{R}$ is said to be **content zero** (resp. **measure zero**) if for any $\epsilon > 0$,

there exists a collection of intervals $\{I_i\}_{i=1}^k$ for some $k \in \mathbb{N}$ (resp. $k = \text{countably infinite} = \infty$) such that $Z \subseteq \bigcup_{i=1}^k I_i$ and $\sum_{i=1}^k \text{length}(I_i) < \epsilon$.

Notice that content zero $\Rightarrow$ measure zero, and the converse holds if the set is compact.

The set of rationals is measure zero but not content zero.

It is equivalent to require I_i are open.

Theorem 32.7. (continuity except content zero implies integrability)

If $f : [a, b] \rightarrow \mathbb{R}$ is bounded on $[a, b]$ and continuous on $[a, b] \setminus Z$, where Z is content zero, then f is Riemann integrable on $[a, b]$.

Theorem 32.8 (Lebesgue's criterion for Riemann-integrability in one variable).

Let $f : [a, b] \rightarrow \mathbb{R}$ be bounded on $[a, b]$ and let Z be the set of discontinuities of f in $[a, b]$.

Then f is Riemann integrable on $[a, b] \iff Z$ is measure zero.

Theorem 32.9. (identity of two integrals)

Let $f, g : [a, b] \rightarrow \mathbb{R}$ be Riemann integrable on $[a, b]$

and $f(x) = g(x)$ for all except finitely many $x \in [a, b]$.

Then $\int_a^b f(x) dx = \int_a^b g(x) dx$.

Example 32.10.

The set of rational numbers in $[0, 1]$ is measure zero set but is not content zero.

Note that the characteristic function of $\mathbb{Q}$ is discontinuous at every point, not merely at rational points, and the Thomae function is continuous on irrationals except rationals.

The complement of a measure-zero set is a dense subset of the universal set.

Theorem 32.11 (compose of integrability, advanced exercise).

Let $f : [a, b] \rightarrow \mathbb{R}$ be a function on $[a, b]$ and $g : [c, d] \rightarrow \mathbb{R}$ be a function on $[c, d]$ with its range in $[a, b]$,

1. If f is continuous on $[a, b]$ and g is Riemann integrable on $[c, d]$, then $f \circ g$ is Riemann integrable on $[c, d]$.

2. It is possible that $f \circ g$ is not Riemann integrable when f is Riemann integrable on $[a, b]$ and g is continuous on $[c, d]$.

33 The fundamental theorem of calculus and the generalized mean value theorem for integrals

Theorem 33.1 (the fundamental theorem of calculus).

1. Let $f : [a, b] \rightarrow \mathbb{R}$ be Riemann integrable on $[a, b]$,
then the function $x \mapsto \int_a^x f(t)dt$ is uniformly continuous on $[a, b]$
and is differentiable at the points at which f is continuous;
in this case, $\frac{d}{dx} \left(\int_a^x f(t)dt \right) \Big|_{x=x_0} = f(x_0)$, for all x_0 at which f is continuous.
2. Let $F : [a, b] \rightarrow \mathbb{R}$ be continuous on $[a, b]$
and is differentiable for all except perhaps finitely many points in $[a, b]$.
Let $f : [a, b] \rightarrow \mathbb{R}$ be a function such that $f(x) = F'(x)$ for those points x where F is differentiable.
In this case, we call the function F an **antiderivative** of f on $[a, b]$.
If f is Riemann integrable on $[a, b]$, then $\int_a^b f(t)dt = F(b) - F(a)$.
(Note that (i) in particular, if F is C^1 on $[a, b]$, then $\int_a^b F'(t)dt = F(b) - F(a)$;
(ii) the “finitely many points” can not be replaced by a set of measure zero, refer to the following theorem.)

Theorem 33.2.

There exists a continuous function $F : [0, 1] \rightarrow \mathbb{R}$
such that $F(a) \neq F(b)$, and $F'(x)$ exists and is equal to zero for all $x \in [0, 1]$ except a set of measure zero;
consider the Cantor function.

Theorem 33.3.

Let $f : [a, b] \rightarrow \mathbb{R}$ be Riemann integrable on $[a, b]$.
Then for any $\epsilon > 0$, there exists $\delta > 0$ such that if P is a partition of $[a, b]$ with $\|P\| < \delta$,
then $0 \leq \max\{|\int_a^b f(x)dx - R_f(P)|, |\int_a^b f(x)dx - L_f(P)|, U_f(P) - \int_a^b f(x)dx, U_f(P) - L_f(P)\} < \epsilon$,
where $R_f(P)$ is a Riemann sum of f at P with arbitrary sample points.

In fact, the above one involving $R_f(P)$ implies f is bounded and
so it is equivalent to Riemann integrability of f on $[a, b]$,
the so-called equivalence of Riemann and Darboux integrals.

Corollary 33.4. (efficient computation for Riemann integrals)

If $f : [a, b] \rightarrow \mathbb{R}$ is Riemann integrable on $[a, b]$,

$$\text{then } \int_a^b f(x)dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(a + i \frac{b-a}{n}) \cdot \frac{b-a}{n}.$$

Theorem 33.5 (the generalized mean value theorem for integrals in one variable).

Let f and g be continuous real-valued functions on $[a, b]$ such that $g(x) \geq 0$ for all $x \in [a, b]$.

Then there exists $c \in [a, b]$ such that $\int_a^b f(x)g(x)dx = f(c) \cdot \int_a^b g(x)dx$.

The number $f(c)$ is called the **g -weighted average of f on $[a, b]$** .

The special case when $g(x) = 1$ is referred to as the **mean value theorem for integrals**.

Theorem 33.6 (integration by parts).

Let f and g be C^1 real-valued functions on $[a, b]$.

$$\text{Then } \int_a^b f(x)g'(x)dx = f(x)g(x) \Big|_{x=a}^b - \int_a^b g(x)f'(x)dx.$$

Theorem 33.7 (strict comparison property).

Let $f, g : [a, b] \rightarrow \mathbb{R}$ be Riemann integrable on $[a, b]$.

If $f(x) < g(x)$ for all $x \in [a, b]$, then $\int_a^b f(x)dx < \int_a^b g(x)dx$.

Theorem 33.8.

Any monotonic real-valued function on $[a, b]$ has at most countable discontinuity
and is differentiable **almost everywhere** (i.e., its complement has measure zero);
the latter one is called **Lebesgue's differentiation theorem**.

Proposition 33.9 (advanced exercise).

Let $f : [0, 1] \rightarrow \mathbb{R}$ be continuous.

$$\text{Then } \lim_{p \rightarrow \infty} \left(\int_0^1 |f(x)|^p dx \right)^{\frac{1}{p}} = \max\{|f(x)| : x \in [0, 1]\}.$$

34 Riemann integrals in several variables

Definition 34.1. (Riemann integral in two variables on a rectangle)

Let $D = [a, b] \times [c, d] = \{(x, y) \in \mathbb{R}^2 : a \leq x \leq b, c \leq y \leq d\}$ be a compact rectangle in $\mathbb{R}^2$.

Let $f : D \rightarrow \mathbb{R}$ be a bounded function.

Let $P = \{a = x_0 < x_1 < \dots < x_m = b; \quad c = y_0 < \dots < y_n = d\}$ be a partition of D .

For $1 \leq i \leq m$ and $1 \leq j \leq n$, let $M_{ij} = \sup\{f(x, y) : x_{i-1} \leq x \leq x_i, y_{j-1} \leq y \leq y_j\}$

and $m_{ij} = \inf\{f(x, y) : x_{i-1} \leq x \leq x_i, y_{j-1} \leq y \leq y_j\}$.

The **lower** and **upper Riemann sums** of f at P are respectively defined

to be $L_f(P) = \sum_{i=1}^m \sum_{j=1}^n m_{ij}(x_i - x_{i-1})(y_j - y_{j-1})$

and $U_f(P) = \sum_{i=1}^m \sum_{j=1}^n M_{ij}(x_i - x_{i-1})(y_j - y_{j-1})$.

The **lower** and **upper Riemann integrals** of f on D are respectively defined

to be $\underline{\int_D} f dA = \sup_{P \text{ is a partition of } D} L_f(P)$ and $\overline{\int_D} f dA = \inf_{P \text{ is a partition of } D} U_f(P)$.

If the lower and upper Riemann integrals of f on D coincide,

we call the common value the **Riemann integral** of f on D ,

denote it by $\int_D f dA$, and say that f is **Riemann integrable** on D .

Definition 34.2. (Riemann integral in several variables on a bounded region)

Let $S \subseteq \mathbb{R}^2$ be bounded and $f : S \rightarrow \mathbb{R}$ be bounded.

Let $D \subseteq \mathbb{R}^2$ be a compact rectangle such that $S \subseteq D$.

The function $\chi_S : \mathbb{R}^2 \rightarrow \{0, 1\}$ defined by $\chi_S(x, y) = \begin{cases} 1, & \text{if } (x, y) \in S, \\ 0, & \text{if } (x, y) \in \mathbb{R}^2 \setminus S, \end{cases}$

is called the **characteristic function** χ_S of S .

If $f\chi_S(x, y) := \begin{cases} f(x, y)\chi_S(x, y), & \text{if } (x, y) \in S, \\ 0, & \text{if } (x, y) \in D \setminus S, \end{cases}$ called the **characteristic extension** of f on S to D ,

is Riemann integrable on D ,

we say that f is **Riemann integrable** on S and define $\int_S f dA = \int_D f\chi_S dA$,

Notice that the definition does not depend on the choice of $D \supset S$.

Similarly, we can define a bounded function $f : S \rightarrow \mathbb{R}$

to be **Riemann integrable** on a bounded set $S \subseteq \mathbb{R}^k$

and use the notation $\int_S f dV^k$, $\int_S f(x) d^k x$, or $\int_S f(x) dx$.

Theorem 34.3.

Let $S \subseteq \mathbb{R}^2$ be bounded and $f, g : S \rightarrow \mathbb{R}$ be Riemann integrable on S . Then:

1. If $c_1, c_2 \in \mathbb{R}$ be constants, then $c_1 f + c_2 g$ is Riemann integrable on S ,
and $\int_S (c_1 f + c_2 g) dA = c_1 \int_S f dA + c_2 \int_S g dA$.
2. If $f(x, y) \leq g(x, y)$ for all $(x, y) \in S$, then $\int_S f dA \leq \int_S g dA$.
3. $|f|$ is Riemann integrable on S and $|\int_S f dA| \leq \int_S |f| dA$.
4. If $f(x, y) \geq 0$ for all $(x, y) \in S$,
then $\int_S f dA$ = the volume of the solid in the (x, y, z) -space,
which is bounded above by the surface $z = f(x, y)$ and bounded below by the region S .
5. Let $S_1, S_2 \subseteq \mathbb{R}^2$ be bounded with $S_1 \cap S_2 = \emptyset$ and $f : S_1 \cup S_2 \rightarrow \mathbb{R}$ be bounded.
Then f is Riemann integrable on S_1 and on S_2
 $\implies f$ is Riemann integrable on $S_1 \cup S_2$;
in which case, $\int_{S_1 \cup S_2} f dA = \int_{S_1} f dA + \int_{S_2} f dA$.
The converse is not true.

Theorem 34.4. (for exercise)

The results similar to Theorems 32.1 and 33.3 work well in several variables.

35 Jordan measurable sets and Riemann integrability

Definition 35.1. (content zero and Jordan measurable sets)

A bounded set $Z \subseteq \mathbb{R}^k$ is said to be **content zero** (respectively, **measure zero**)

if for any $\epsilon > 0$ there exists a finite (resp. countable) collection of k -dimensional open rectangles $D_1, \dots, D_n$ (resp. $n = \infty$) such that $Z \subseteq \cup_{i=1}^n D_i$ and $\sum_{i=1}^n \text{volume}(D_i) < \epsilon$.

A bounded set $S \subseteq \mathbb{R}^k$ is said to be **Jordan** (respectively, **Riemann**) **measurable**

if its boundary ∂S is content (resp. measure) zero;

is fact, they are equivalent since in $\mathbb{R}^k$ the boundary of a bounded set is compact.

The set of rational numbers and the fat Cantor set are not Jordan measurable, while the Cantor set is Jordan measurable.

In fact, for the Cantor set, it is YES for content/measure zero, Jordan/Riemann measurable, length-definable, but for the fat Cantor set, it NO for all, for Q it is YES only for measure zero.)

Theorem 35.2. (continuity except content zero implies Riemann integrability on a rectangle)

Let $D \subseteq \mathbb{R}^k$ be a compact rectangle and $f : D \rightarrow \mathbb{R}$ be bounded.

If f is continuous on $D \setminus Z$, where $Z \subseteq D$ is content zero or measure zero, then f is Riemann integrable on D .

Theorem 35.3. (sufficient conditions for content zero)

1. A set of content zero is measure zero, but not the converse, e.g. Q .
 A compact set of measure zero is content zero.
 A subset of a set with content zero and measure zero is content zero and measure zero, respectively.
 A finite union of sets with content zero is content zero.
 A countable union of sets with content zero might not be content zero, e.g. Q .
 A countable union of sets with measure zero is measure zero.
2. Let $f : (a, b) \rightarrow \mathbb{R}^2$ be a C^1 function.
 Then $f([c, d])$ is content zero and is measure zero whenever $a < c < d < b$.
 Here C^1 cannot be replaced by C^0 ; consider the Peano curve, which is in $C^0(\mathbb{R}, \mathbb{R}^2)$
 For $C^0(\mathbb{R}, \mathbb{R})$, consider the Cantor function restricted to the Cantor set of measure zero.
3. Let $f : \mathbb{R}^k \rightarrow \mathbb{R}^n$ be C^1 and $k < n$.
 If S is a compact subset $S \subseteq \mathbb{R}^k$, then $f(S)$ is content zero and measure zero.

Theorem 35.4. (Lebesgue's criterion for Riemann integrability in several variables, cf. Theorem 32.8.)

Let $S \subseteq \mathbb{R}^k$ be bounded and $f : S \rightarrow \mathbb{R}$ be bounded, and let Z be the set of discontinuities of f in S . Then

1. f is Riemann integrable on $S \implies Z$ is measure zero;
 notice that ∂S might not be measure zero, for instance the zero function on $Q \cap [0, 1]$.
2. f is Riemann integrable on $S \iff Z \cup \partial S$ is measure zero.
3. If ∂S is measure zero, then f is Riemann integrable on $S \iff Z$ is measure zero.
4. Let D be a compact rectangle in $\mathbb{R}^k$ such that $S \subseteq D$.
 Then f is Riemann integrable on $S \iff f\chi_S$ is continuous on $D \setminus Z_1$, where $Z_1 \subseteq D$ is measure zero;
 for the latter one, it is often said that f is continuous **almost everywhere**.

Corollary 35.5.

1. If $f : \mathbb{R}^k \rightarrow \mathbb{R}$ is Riemann integrable on S and on T , and $S \cap T$ is content zero, then f is Riemann integrable on $S \cup T$, and $\int_{S \cup T} f dV^k = \int_S f dV^k + \int_T f dV^k$.
2. Let f be Riemann integrable on $S \subseteq \mathbb{R}^k$ and $g : \mathbb{R}^k \rightarrow \mathbb{R}$ be bounded.
 If $g(x) = f(x)$ for all $x \in S \setminus T$, where T is content zero, then g is Riemann integrable on S and $\int_S g dV^k = \int_S f dV^k$.

Remark 35.6. (inner and outer volumes)

Let $S \subseteq \mathbb{R}^k$ be bounded. Let D be a compact rectangle such that $S \subseteq D$, and let P a partition of D .

Then the lower Riemann sum $L_{\chi_S}(P)$ (resp. the upper Riemann sum $U_{\chi_S}(P)$) is

the sum of the areas of the sub-rectangles in P which is contained in S (resp. intersect S).

We call the lower Riemann integral $\int_D \chi_S dV^k$ the **inner volume** of S

and call the upper Riemann integral $\int_D \chi_S dV^k$ the **outer volume** of S .

Then S is Jordan measurable $\iff S$ is Riemann measurable $\iff \chi_S$ is Riemann integrable on S ,

and in this case, we define the (Jordan) volume of S to be $\text{volume}(S) := \int_S 1 dV^k = \int_D \chi_S dV^k$.

Notice that the inner area and the outer area are distinct for the set of rationals and the fat Cantor set, but are the same for the Cantor set.

36 The generalized mean value theorem and the second mean value theorem for integrals

Proposition 36.1. (zero Riemann integral)

Let $S \subseteq \mathbb{R}^k$ be bounded and $f : S \rightarrow \mathbb{R}$ be bounded. Then

1. If S is content zero, then f is Riemann integrable on S and $\int_S f dV^k = 0$.
2. If S is measure zero and f is Riemann integrable on S , then $\int_S f dV^k = 0$.
3. Let $S = Q \cap [0, 1]$ and $f = \chi_S$, then S is measure zero and f is not Riemann integrable.
4. If $f(x) \geq 0$ for all $x \in S$ and $\int_S f dV^k = 0$, then $\{x \in S : f(x) \neq 0\}$ is measure zero.

Corollary 36.2.

1. If $f : \mathbb{R}^k \rightarrow \mathbb{R}$ is Riemann integrable on S and on T , and $S \cap T$ is content zero, then f is Riemann integrable on $S \cup T$, and $\int_{S \cup T} f dV^k = \int_S f dV^k + \int_T f dV^k$.
2. Let f be Riemann integrable on $S \subseteq \mathbb{R}^k$ and $g : \mathbb{R}^k \rightarrow \mathbb{R}$ be bounded. If $g(x) = f(x)$ for all $x \in S \setminus T$, where T is content zero, then g is Riemann integrable on S and $\int_S g dV^k = \int_S f dV^k$.

Theorem 36.3 (the generalized mean value theorem for integrals in several variables).

Let $S \subseteq \mathbb{R}^k$ be a compact, connected, and Jordan measurable set.

If $f, g : S \rightarrow \mathbb{R}$ are continuous on S with $g \geq 0$,

then there exists $c \in S$ such that $\int_S f(x)g(x) d^k x = f(c) \int_S g(x) d^k x$.

In particular, if $f : S \rightarrow \mathbb{R}$ is continuous on S ,

then there exists $c \in S$ such that $\int_S f(x) d^k x = f(c) \cdot \text{volume}(S)$.

Theorem 36.4 (the second mean value theorem for integrals).

Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous on $[a, b]$ and $g : [a, b] \rightarrow \mathbb{R}$ be C^1 and increasing on $[a, b]$.

Then there exists $c \in [a, b]$ such that $\int_a^b f(x)g(x)dx = g(a) \int_a^c f(x)dx + g(b) \int_c^b f(x)dx$.

About g , one can only assume that g is monotone, but the proof is much harder.

Theorem 36.5 (the C^1 mean value theorem for vector-valued functions, for advanced exercise).

Let $S \subseteq \mathbb{R}^k$ be open and $f : S \rightarrow \mathbb{R}^k$ be C^1 on S .

Let $x, h \in \mathbb{R}^k$ be vectors such that $\{x + th : 0 \leq t \leq 1\} \subseteq S$.

Then $f(x + h) - f(x) = \left(\int_0^1 Df(x + th) dt \right) \cdot h$,

where Df denotes the Jacobian matrix of f and $\cdot$ denotes the matrix multiplication.

37 Iterated integrals and Fubini's theorem

Theorem 37.1 (Fubini's theorem I).

Let $D = [a, b] \times [c, d]$ be a closed rectangle in $\mathbb{R}^2$ and $f : D \rightarrow \mathbb{R}$ be Riemann integrable.

1. If for any $y \in [c, d]$, the function $g(x) := f(x, y)$ is a Riemann integrable on $[a, b]$, and the function $h(y) := \int_a^b f(x, y) dx$ is Riemann integrable on $[c, d]$, then
$$\iint_D f dA = \int_c^d \left[\int_a^b f(x, y) dx \right] dy.$$
2. If for any $x \in [a, b]$, $f(x, y)$ is a Riemann integrable function of y on $[c, d]$, and the function $\int_c^d f(x, y) dy$ is a Riemann integrable function of x on $[a, b]$, then
$$\iint_D f dA = \int_a^b \left[\int_c^d f(x, y) dy \right] dx.$$
3.
$$\iint_D f dA = \int_c^d \left[\int_a^b f(x, y) dx \right] dy = \int_c^d \left[\int_a^b f(x, y) dx \right] dy = \int_a^b \left[\int_c^d f(x, y) dy \right] dx = \int_a^b \left[\int_c^d f(x, y) dy \right] dx.$$
4. It is possible that the function $y \mapsto \int_a^b f(x, y) dx$ is not defined on a set of measure zero, there in, for each y , one has $\int_a^b f(x, y) dx < \int_a^b f(x, y) dx$.

The integrals on the right hand side of the above equations are called **iterated integrals** of f on D .

Theorem 37.2 (Fubini's theorem II).

1. Let $D \subseteq \mathbb{R}^2$ be a region which is bounded above and below by two C^1 functions, namely $D = \{(x, y) \in \mathbb{R}^2 : a \leq x \leq b, g_1(x) \leq y \leq g_2(x)\}$, where $g_1 \leq g_2$ are C^1 on $[a, b]$. If $f : D \rightarrow \mathbb{R}$ is continuous on D , then

$$\iint_D f dA = \int_{x=a}^b \int_{y=g_1(x)}^{g_2(x)} f(x, y) dy dx.$$

2. Let $D = \{(x, y, z) \in \mathbb{R}^3 : a \leq x \leq b, g_1(x) \leq y \leq g_2(x), h_1(x, y) \leq z \leq h_2(x, y)\}$, where $g_1 \leq g_2$ are C^1 on $[a, b]$ and $h_1 \leq h_2$ are C^1 on $\{(x, y) \in \mathbb{R}^2 : a \leq x \leq b, g_1(x) \leq y \leq g_2(x)\}$. If $f : D \rightarrow \mathbb{R}$ is Riemann integrable on D , then
$$\iiint_D f dV = \int_{x=a}^b \int_{y=g_1(x)}^{g_2(x)} \int_{z=h_1(x, y)}^{h_2(x, y)} f(x, y, z) dz dy dx.$$

Remark 37.3.

1. If $f(x, y) \geq 0$ for all $(x, y) \in D$, then $\iint_D f dA$ is the **volume** of the region in $\mathbb{R}^3$ under the surface $z = f(x, y)$ and above the region D in $\mathbb{R}^2$.
2. Let $D \subseteq \mathbb{R}^3$ be a region. If $f(x, y, z)$ is a **density function** which describes the distribution of a substance throughout D , then $\iiint_D f dV =$ the **quantity** of the substance; for instance, the mass of the object.
3. If a massive object with mass density $f(x, y, z)$ occupies the region $D \subseteq \mathbb{R}^3$, then its total **mass** is $\iiint_D f dV$, the **center of gravity** of this object is the point $(\frac{\iiint_D x f(x, y, z) dV}{\iiint_D f dV}, \frac{\iiint_D y f(x, y, z) dV}{\iiint_D f dV}, \frac{\iiint_D z f(x, y, z) dV}{\iiint_D f dV})$, and the **centroid of the region** D is $(\frac{\iiint_D x dV}{\iiint_D 1 dV}, \frac{\iiint_D y dV}{\iiint_D 1 dV}, \frac{\iiint_D z dV}{\iiint_D 1 dV})$ (independent of f). Let ℓ be a line in $\mathbb{R}^3$, then the **moment of inertia** of the object about the line ℓ is $\iiint_D [d(x, y, z)]^2 f(x, y, z) dV$, where $d(x, y, z)$ is the distance from the point $(x, y, z) \in D$ to the line ℓ .

38 Change of variables for Riemann integrals

Theorem 38.1. (change of variables for Riemann integrals in one variable, the so-called substitution)

Let $g : [a, b] \mapsto \mathbb{R}$ be C^1 and $f : g([a, b]) \mapsto \mathbb{R}$ be continuous.

Then $\int_{g(a)}^{g(b)} f(x)dx = \int_a^b f(g(y))g'(y)dy$.

Theorem 38.2. (change of variables for Riemann integrals in several variables)

Let U and V be open subsets of $\mathbb{R}^k$ and $g : U \rightarrow V$ be one-to-one and C^1 on U such that $Dg(y)$ is invertible for all $y \in U$.

Let $S \subseteq V$ be Jordan measurable such that $g^{-1}(S)$ is Jordan measurable.

1. If $f : S \rightarrow \mathbb{R}$ is Riemann integrable on S ,
then $f \circ g$ is Riemann integrable on $g^{-1}(S)$,
and $\int_S f(x)d^k x = \int_{g^{-1}(S)} f(g(y))|det Dg(y)|d^k y$.
2. In particular, if g is transformation on $\mathbb{R}^k$
with an invertible matrix representation $g(y) = Ay$,
then $\int_S f(x)d^k x = |det A| \int_{A^{-1}(S)} f(Ay)d^k y$.

Corollary 38.3. (special coordinates)

Assume the hypotheses of the previous theorem hold.

1. Let $k = 1$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ be one-to-one and C^1 on $\mathbb{R}$.
Then $\int_S f(x)dx = \int_{g^{-1}(S)} f(g(y))|g'(y)|dy$,
where $\int_S$ and $\int_{g^{-1}(S)}$ always mean the integral from the left to the right.
2. (polar coordinates)
Let $g : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be defined by $g(r, \theta) = (r \cos \theta, r \sin \theta)$
for $r \geq 0$ and $0 \leq \theta < 2\pi$. Then $\iint_S f(x, y)dA = \iint_{g^{-1}(S)} f(r \cos \theta, r \sin \theta)rdrd\theta$.
3. (cylindrical coordinates)
Let $g : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be defined by $g(r, \theta, z) = (r \cos \theta, r \sin \theta, z)$
for $r \geq 0$, $0 \leq \theta < 2\pi$ and $z \in \mathbb{R}$.
Then $\iiint_S f dV = \iiint_{g^{-1}(S)} f(r \cos \theta, r \sin \theta, z)rdrd\theta dz$.
4. (spherical coordinates)
Let $g : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be defined by $g(r, \theta, \varphi) = (r \sin \varphi \cos \theta, r \sin \varphi \sin \theta, r \cos \varphi)$
for $r \geq 0$, $0 \leq \theta < 2\pi$ and $0 \leq \varphi < \pi$.
Then $\iiint_S f dV = \iiint_{g^{-1}(S)} f(r \sin \varphi \cos \theta, r \sin \varphi \sin \theta, r \cos \varphi)r^2 \sin \varphi drd\theta d\varphi$.
5. The volume $vol(B_r^n)$ of the n -dimensional ball of radius $r > 0$ is $r^n vol(B_1^n) = \frac{2\pi r^n}{n} vol(B_1^{n-2})$ for $n \geq 3$.

39 Functions defined by integrals and the bounded convergence theorem

Theorem 39.1. (*partial derivatives and Riemann integrals*)

Let $S \subseteq \mathbb{R}^k$ be open, $T \subseteq \mathbb{R}^m$ be compact and Jordan measurable, and $S \times T = \{(x, y) \in \mathbb{R}^{k+m} : x \in S, y \in T\}$.

1. If $f : S \times T \rightarrow \mathbb{R}$ be continuous on $S \times T$, then the function $x \mapsto \int_T f(x, y) d^m y$ is continuous on S .
2. If, in addition, $\frac{\partial f}{\partial x_i}(x, y)$ is continuous on $S \times T$ for $1 \leq i \leq k$, then the function $x \mapsto \int_T f(x, y) d^m y$ is C^1 on S and for each $x = (x_1, \dots, x_k) \in S$ and each $1 \leq i \leq k$,

$$\frac{\partial}{\partial x_i} \int_T f(x, y) d^m y = \int_T \frac{\partial f}{\partial x_i}(x, y) d^m y.$$

Corollary 39.2.

Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be continuous on $\mathbb{R}^2$ such that $\frac{\partial f}{\partial x}(x, y)$ is continuous on $\mathbb{R}^2$ and let $g : \mathbb{R} \rightarrow \mathbb{R}$ be C^1 on $\mathbb{R}$.

Then $\frac{d}{dx} \int_a^{g(x)} f(x, y) dy = f(x, g(x))g'(x) + \int_a^{g(x)} \frac{\partial f}{\partial x}(x, y) dy$.

Theorem 39.3 (the bounded convergence theorem).

Let $T \subseteq \mathbb{R}^m$ be Jordan measurable

and $\{f_n : T \rightarrow \mathbb{R}\}_{n=1}^\infty$ be a sequence of Riemann integrable functions on T .

Assume that $\lim_{n \rightarrow \infty} f_n(y) = f(y)$ for each $y \in T$,

where f is a Riemann integrable function on T ,

and that there exists a constant $M > 0$ such that $|f_n(y)| \leq M$ for all $y \in T$ and for all $n \in \mathbb{N}$.

Then $\lim_{n \rightarrow \infty} \int_T f_n(y) d^m y = \int_T f(y) d^m y$.

For the Lebesgue integral, the integrability of the limit function f is a conclusion instead of a condition; refer to the so-called dominated convergence theorem.

One can consider the characteristic function of rationals as counter-example for Riemann integrability.

Corollary 39.4.

Let $S \subseteq \mathbb{R}^k$ be a set, $T \subseteq \mathbb{R}^m$ be a Jordan measurable set,

and $S \times T = \{(x, y) \in \mathbb{R}^{k+m} : x \in S, y \in T\}$.

Let $f : S \times T \rightarrow \mathbb{R}$ be Riemann integrable on T for each $x \in S$.

1. If for each $y \in T$ the function $x \mapsto f(x, y)$ is continuous on S and there exists a constant $M > 0$ such that $|f(x, y)| \leq M$ for all $(x, y) \in S \times T$, then the function $x \mapsto \int_T f(x, y) d^m y$ is continuous on S .
2. If S is open, for each $y \in T$ the function $x \mapsto f(x, y)$ is C^1 on S , for each $x \in S$, the function $y \mapsto \frac{\partial f}{\partial x_i}(x, y)$ is Riemann integrable on T , and there exists a constant $M > 0$ such that $\|\nabla_x f(x, y)\| \leq M$ for all $(x, y) \in S \times T$, where $\nabla_x f(x, y)$ denotes given y , the gradient of $f(x, y)$ with respect to x , then the function $x \mapsto \int_T f(x, y) d^m y$ is C^1 on S and for each $x = (x_1, \dots, x_k) \in S$ and each $1 \leq i \leq k$,

$$\frac{\partial}{\partial x_i} \int_T f(x, y) d^m y = \int_T \frac{\partial f}{\partial x_i}(x, y) d^m y.$$

40 Improper integrals of type I

Definition 40.1.

The following kinds of typical integrals are called the *improper integrals of type I*.

1. Let $f : [a, \infty) \rightarrow \mathbb{R}$ be Riemann integrable on $[a, t]$ for all $t \in [a, \infty)$.

We define the **improper integral** by $\int_a^\infty f(x)dx = \lim_{t \rightarrow \infty} \int_a^t f(x)dx$ and

say that the improper integral **converges** (or simply is **convergent**) if the limit exists as a finite number

and say that the improper integral **diverges** (or is **divergent**) otherwise.

Similarly, one can define the **improper integral** $\int_{-\infty}^b f(x)dx$.

2. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function.

If there exists $b \in \mathbb{R}$ such that both improper integrals $\int_{-\infty}^b f(x)dx$ and $\int_b^\infty f(x)dx$ converge as above,

then we say that $\int_{-\infty}^\infty f(x)dx$ **converges** (or is **convergent**)

and write $\int_{-\infty}^\infty f(x)dx = \int_{-\infty}^b f(x)dx + \int_b^\infty f(x)dx = \lim_{t \rightarrow -\infty} \int_t^b f(x)dx + \lim_{s \rightarrow \infty} \int_b^s f(x)dx$.

and say that the improper integral **diverges** (or is **divergent**) otherwise.

Theorem 40.2 (comparison test for improper integrals of type I).

Let $f, g : [a, \infty) \rightarrow \mathbb{R}$ be Riemann integrable on $[a, t]$ for all $t \in [a, \infty)$.

Suppose $0 \leq f(x) \leq g(x)$ for all sufficiently large $x \in [a, \infty)$.

1. If $\int_a^\infty g(x)dx$ converges, then $\int_a^\infty f(x)dx$ converges.
2. If $\int_a^\infty f(x)dx$ diverges, then $\int_a^\infty g(x)dx$ diverges.

Corollary 40.3 (limit comparison test for improper integrals of type I).

Let $f, g : [a, \infty) \rightarrow \mathbb{R}$ be Riemann integrable on $[a, t]$ for all $t \in [a, \infty)$.

Suppose $f(x) > 0$ and $g(x) > 0$ for all $x \in [a, \infty)$ and $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = \ell$.

1. If $0 < \ell < \infty$, then $\int_a^\infty f(x)dx$ and $\int_a^\infty g(x)dx$ both converges or diverges.
2. If $\ell = 0$ and $\int_a^\infty g(x)dx$ converges, then $\int_a^\infty f(x)dx$ converges.
3. If $\ell = \infty$ and $\int_a^\infty g(x)dx$ diverges, then $\int_a^\infty f(x)dx$ diverges.

Corollary 40.4.

Let $f : [a, \infty) \rightarrow \mathbb{R}$ be Riemann integrable on $[a, t]$ for all $t \in [a, \infty)$.

1. If $0 \leq f(x) \leq cx^{-p}$ for all sufficiently large $x \in [a, \infty)$, where $c > 0$ and $p > 1$, then $\int_a^\infty f(x)dx$ converges.
2. If $f(x) \geq cx^{-1}$ for all sufficiently large $x \in [a, \infty)$, where $c > 0$, then $\int_a^\infty f(x)dx$ diverges.

41 Improper integrals of type II

Definition 41.1.

The following kinds of typical integrals are called the *improper integrals of type II*.

1. Let $f : [a, b) \rightarrow \mathbb{R}$ be Riemann integrable on $[a, t]$ for all $t \in [a, b)$ such that $\lim_{x \rightarrow b^-} |f(x)|$ equals to ∞ or does not exist.

We define the **improper integral** $\int_a^b f(x)dx = \lim_{t \rightarrow b^-} \int_a^t f(x)dx$,

and say that the improper integral **converges** (or is **convergent**) if the limit exists as a finite number.

Otherwise, we say that the improper integral **diverges** (or is **divergent**).

Similarly, one can define the **improper integral** $\int_a^b f(x)dx = \lim_{t \rightarrow a^+} \int_t^b f(x)dx$.

2. Let $f : [a, c) \cup (c, b] \rightarrow \mathbb{R}$ be a function such that $\lim_{x \rightarrow c^-} |f(x)|$ equals to ∞ or does not exist, and $\lim_{x \rightarrow c^+} |f(x)|$ equals to ∞ or does not exist.

If both improper integrals $\int_a^c f(x)dx$ and $\int_c^b f(x)dx$ converge,

then we say that the improper integral $\int_a^b f(x)dx$ **converges** (or is **convergent**)

and write $\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx = \lim_{t \rightarrow c^-} \int_a^t f(x)dx + \lim_{s \rightarrow c^+} \int_s^b f(x)dx$.

Otherwise, we say that the improper integral **diverges** (or is **divergent**).

3. Let $f : (a, b) \rightarrow \mathbb{R}$ be a function such that $\lim_{x \rightarrow a^+} |f(x)|$ equals to ∞ or does not exist, and $\lim_{x \rightarrow b^-} |f(x)|$ equals to ∞ or does not exist.

If there exists $c \in (a, b)$ such that both improper integrals $\int_a^c f(x)dx$ and $\int_c^b f(x)dx$ converge as in item 1,

then we say that $\int_a^b f(x)dx$ **converges** (or is **convergent**)

and write $\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx$.

Otherwise, we say that the improper integral **diverges** (or is **divergent**).

Theorem 41.2 (comparison test for improper integrals of type II).

Let $f, g : [a, b) \rightarrow \mathbb{R}$ be Riemann integrable on $[a, t]$ for all $t \in [a, b)$.

Suppose $0 \leq f(x) \leq g(x)$ for all $x \in [a, b)$ sufficiently close to b .

1. If $\lim_{x \rightarrow b^-} g(x) = \infty$ and $\int_a^b g(x)dx$ converges, then $\int_a^b f(x)dx$ converges.
2. If $\lim_{x \rightarrow b^-} f(x) = \infty$ and $\int_a^b f(x)dx$ diverges, then $\int_a^b g(x)dx$ diverges.

Corollary 41.3.

Let $f : (0, b] \rightarrow \mathbb{R}$ be Riemann integrable on $(t, b]$ for all $t \in (0, b]$.

1. If $0 \leq f(x) \leq cx^{-p}$ for all $x \in (0, b]$ near 0, where $c > 0$ and $p < 1$ (not $p > 1$), then $\int_0^b f(x)dx$ converges.
2. If $f(x) \geq cx^{-1}$ for all $x \in (0, b]$ near 0, where $c > 0$, then $\int_0^b f(x)dx$ diverges.

Corollary 41.4.

Let $f : [a, b) \rightarrow \mathbb{R}$ be Riemann integrable on $[a, t]$ for all $t \in [a, b)$.

1. If $0 \leq f(x) \leq c(b-x)^{-p}$ for all $x \in [a, b)$ near b , where $c > 0$ and $p < 1$ (not $p > 1$), then $\int_a^b f(x)dx$ converges.
2. If $f(x) \geq c(b-x)^{-1}$ for all $x \in [a, b)$ near b , where $c > 0$, then $\int_a^b f(x)dx$ diverges.

42 Absolute convergence and Cauchy principle value

Theorem 42.1 (Dirichlet's test for convergence of improper integrals).

Let $f : [a, \infty) \rightarrow \mathbb{R}$ be continuous and $g : [a, \infty) \rightarrow \mathbb{R}$ be C^1 .

Suppose that there exists $M > 0$ such that $|\int_a^x f(t)dt| < M$ for all $x \in [a, \infty)$

and that $g'(x) \leq 0$ for all $x \in [a, \infty)$ and $\lim_{x \rightarrow \infty} g(x) = 0$.

Then $\int_a^\infty f(x)g(x)dx$ converges.

One can replace the smoothness about g by that g is bounded and monotonic.

Definition 42.2.

For $f : [a, \infty) \rightarrow \mathbb{C}$, we define $\int_a^\infty f(x)dx = \int_a^\infty \operatorname{Re} f(x)dx + i \int_a^\infty \operatorname{Im} f(x)dx$ and say it is **convergent**, provided that the two improper integrals both converge; otherwise, we say the improper integral $\int_a^\infty f(x)dx$ is **divergent**. The improper integral $\int_a^\infty f(x)dx$ is said to be **absolutely convergent** if $\int_a^\infty |f(x)|dx$ converges.

Theorem 42.3 (absolute convergence implies convergence).

Let $f : [a, \infty) \rightarrow \mathbb{R}$ (or $\mathbb{C}$) be a function

such that both $|f|$ and f are Riemann integrable on $[a, b]$ for all $b \in [a, \infty)$.

If $\int_a^\infty |f(x)|dx$ converges, then $\int_a^\infty f(x)dx$ converges.

The converse does not hold, for instance $\int_0^\infty \frac{\sin(x)}{x}dx$.

The integrability on finite region is needed; see the counter example $f(x) = 1$ on rationals and $f(x) = -1$ on irrationals.

Definition 42.4 (Cauchy principle value of improper integrals).

1. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be Riemann integrable on $[-t, t]$ for all $t \in \mathbb{R}$.

The **Cauchy principle value** of the improper integral $\int_{-\infty}^\infty f(x)dx$ is defined to be

$$\text{P.V. } \int_{-\infty}^\infty f(x)dx = \lim_{t \rightarrow \infty} \int_{-t}^t f(x)dx, \text{ provided the limit exists as a finite number.}$$

2. Let $f : [a, c) \cup (c, b]$ be a function such that $\lim_{x \rightarrow c} |f(x)| = \infty$

and that f is Riemann integrable on $[a, c - \epsilon]$ and on $[c + \epsilon, b]$ for all $\epsilon > 0$.

The **Cauchy principle value** of the improper integral $\int_a^b f(x)dx$ is defined to be

$$\text{P.V. } \int_a^b f(x)dx = \lim_{\epsilon \rightarrow 0^+} \left[\int_a^{c-\epsilon} f(x)dx + \int_{c+\epsilon}^b f(x)dx \right],$$

provided the limit exists as a finite number.

Remark 42.5.

The existence of the Cauchy principle value of an improper integral is guaranteed by the convergence of the improper integral; but not the other way around.

Theorem 42.6.

Let $a < 0 < b$ and $f : [a, b] \rightarrow \mathbb{R}$ be continuous on $[a, b]$ and differentiable at 0,

then P.V. $\int_a^b \frac{f(x)}{x}dx$ exists.

43 Improper integrals in $\mathbb{R}^k$

Theorem 43.1.

Let $S \subseteq \mathbb{R}^k$ be open and $f: S \rightarrow \mathbb{R}$ such that $f(x) \geq 0$ for all $x \in S$.

Let $\{K_i\}_{i=1}^\infty$ and $\{G_i\}_{i=1}^\infty$ be sequences of compact subsets of S

such that $K_i \subseteq K_{i+1}$ and $G_i \subseteq G_{i+1}$ for all $i \geq 1$ and $\bigcup_{i=1}^\infty K_i^{int} = S = \bigcup_{i=1}^\infty G_i^{int}$

and f is Riemann integrable on each K_i and G_i .

Then $\lim_{i \rightarrow \infty} \int_{K_i} f(x) d^k x = \lim_{i \rightarrow \infty} \int_{G_i} f(x) d^k x$,

where the limits may be finite or $+\infty$.

Definition 43.2. (improper integral in $\mathbb{R}^k$)

Let $S \subseteq \mathbb{R}^k$ be open and $f: S \rightarrow \mathbb{R}$

such that either S is unbounded or f is unbounded on S .

If $f: S \rightarrow [0, \infty)$,

we define the **improper integral** $\int_S f(x) d^k x = \lim_{i \rightarrow \infty} \int_{K_i} f(x) d^k x$,

where $\{K_i\}_{i=1}^\infty$ is a sequence of compact subsets of S

such that f is Riemann integrable on K_i and $K_i \subseteq K_{i+1}$ for all $i \geq 1$, and $\bigcup_{i=1}^\infty K_i^{int} = S$.

We say the **improper integral** $\int_S f(x) d^k x$ **converges**

if the limit $\lim_{i \rightarrow \infty} \int_{K_i} f(x) d^k x$ exists as a finite number

and say that the **improper integral** $\int_S f(x) d^k x$ **diverges** otherwise.

If $f: S \rightarrow \mathbb{R}$,

we denote $f^+(x) = \max\{f(x), 0\}$ and $f^-(x) = \max\{-f(x), 0\}$

and define the improper integral $\int_S f(x) d^k x = \int_S f^+(x) d^k x - \int_S f^-(x) d^k x$,

provided that the improper integrals on the right hand side are not both $+\infty$.

Similarly, we can define the **convergence and divergence of the improper integral**.

(If the improper integral $\int_S f(x) d^k x$ is well defined and is convergent, Fubini's theorem for improper integral will hold.)

If $f: S \rightarrow \mathbb{C}$, we simply consider its real and imaginary parts separately.

Theorem 43.3.

Let $a > 0$ and $p > 0$.

1. The improper integral $\int_{\{x \in \mathbb{R}^k: \|x\| < a\}} \frac{1}{\|x\|^p} d^k x$ converges $\iff p < k$.
2. The improper integral $\int_{\{x \in \mathbb{R}^k: \|x\| > a\}} \frac{1}{\|x\|^p} d^k x$ converges $\iff p > k$.

Theorem 43.4. $\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$.

44 The Lebesgue measure and Lebesgue measurable sets

Definition 44.1.

A set $S \subseteq \mathbb{R}^2$ said to be **tiled**

if it is the union of a finite collection of rectangles with disjoint interiors.

Definition 44.2.

The **Lebesgue measure** $m(S) \in [0, \infty]$ of a suitable set $S \subseteq \mathbb{R}^2$ is defined as follows:

1. if S is tiled, then $m(S) := \sum_{i=1}^n \text{area}(D_i)$,
where $S = \cup_{i=1}^n D_i$ and the D_i 's are rectangles with disjoint interiors;
2. if S is compact, then $m(S) := \inf\{m(T) : T \text{ is tiled and } S \subseteq T\}$;
3. if S is open, then $m(S) := \sup\{m(T) : T \text{ is tiled and } T \subseteq S\}$;
4. if $S \subseteq \mathbb{R}^2$ satisfies
 $\sup\{m(T) : T \text{ is compact and } T \subseteq S\} = \inf\{m(T) : T \text{ is open and } S \subseteq T\}$,
then $m(S) :=$ the common value of the above equality;

in which case we say that S is **Lebesgue measurable**.

Similarly one can define the **Lebesgue measure** of a Lebesgue measurable set $S \subseteq \mathbb{R}^k$.

Remark 44.3.

1. Every open set, every closed set, every intersection of a sequence of open sets, and every union of a sequence of closed sets, are all Lebesgue measurable.
2. If $\{S_i\}_{i=1}^\infty$ is a sequence of disjoint Lebesgue measurable sets in $\mathbb{R}^k$, then $\cup_{i=1}^\infty S_i$ is Lebesgue measurable and $m(\cup_{i=1}^\infty S_i) = \sum_{i=1}^\infty m(S_i)$.
3. A Lebesgue measurable set $S \subseteq \mathbb{R}^k$ has Lebesgue measure zero
 $\iff$ for any $\epsilon > 0$ there exists a sequence of k -dimensional rectangular boxes $\{D_i\}_{i=1}^\infty$ such that $S \subseteq \cup_{i=1}^\infty D_i$ and $\sum_{i=1}^\infty \text{volume}(D_i) < \epsilon$.
4. The set of all rationals in $[0, 1]$ is a Lebesgue measurable set with Lebesgue measure zero, but is not Jordan measurable.
5. The Cantor set is Lebesgue measure zero and contains no nontrivial intervals and is **nowhere dense** (i.e., the interior of the closure of the set is empty).

45 The Lebesgue integral

Definition 45.1.

1. A function $f : \mathbb{R}^k \rightarrow \mathbb{R}$ is said to be **Lebesgue measurable** if for every interval $I \subseteq \mathbb{R}$, the set $f^{-1}(I) = \{x \in \mathbb{R}^k : f(x) \in I\}$ is Lebesgue measurable, i.e., $m(f^{-1}(I))$ is well defined.
2. For a Lebesgue measurable function $f : \mathbb{R}^k \rightarrow [0, \infty)$, we consider $[0, \infty) = \cup_{i=0}^{\infty} [\frac{i}{2^n}, \frac{i+1}{2^n})$, denote $S_n(f) = \sum_{i=0}^{\infty} \frac{i}{2^n} m(f^{-1}([\frac{i}{2^n}, \frac{i+1}{2^n})))$ for $n \in \mathbb{N}$, and define the **Lebesgue integral** of f on $\mathbb{R}^k$ to be $\int_{\mathbb{R}^k} f dm = \lim_{n \rightarrow \infty} S_n(f)$, possibly $+\infty$.
3. For a Lebesgue measurable function $f : \mathbb{R}^k \rightarrow [0, \infty)$ and a Lebesgue measurable set $S \subseteq \mathbb{R}^k$, the **Lebesgue integral** of f on S is defined to be $\int_S f dm = \int_{\mathbb{R}^k} f \chi_S dm$.
4. For a Lebesgue measurable function $f : \mathbb{R}^k \rightarrow \mathbb{R}$ and a Lebesgue measurable set $S \subseteq \mathbb{R}^k$, the **Lebesgue integral** of f on S is defined to be $\int_S f dm = \int_S f^+ dm - \int_S f^- dm$, provided the integrals on the right hand side are not both $+\infty$, where $f^+(x) = \max\{f(x), 0\}$ and $f^-(x) = \max\{-f(x), 0\}$.

Remark 45.2.

1. Every continuous function is Lebesgue measurable.
2. If f is Riemann integrable on S , then $f \chi_S$ is Lebesgue measurable.
3. If $\{f_i\}_{i=1}^{\infty}$ is a sequence of Lebesgue measurable functions and f is a function such that for each x , we have $f_i(x) \rightarrow f(x)$ as $i \rightarrow \infty$, then the function f is Lebesgue measurable.
4. If the Riemann integral $\int_S f dV^k$ exists, then the Lebesgue integral $\int_S f dm$ exists and the two are equal.
5. Let $f : \mathbb{R}^k \rightarrow \mathbb{R}$ be a function and $S \subseteq \mathbb{R}^k$ be bounded. Then f is Riemann integrable on S $\iff$ f is bounded on S and the set of points in $\mathbb{R}^k$ at which $f \chi_S$ is discontinuous has Lebesgue measure zero.
6. Let S be the set of all rationals in $[0, 1]$. Then χ_S is Lebesgue measurable and the Lebesgue integral of χ_S is zero, but χ_S is not Riemann integrable.

46 Definition of definite Riemann-Stieltjes integral

Definition 46.1. (*Riemann-Stieltjes sum and integrability*)

Let $f, g : [a, b] \rightarrow \mathbb{R}$ be bounded functions

Let $P = \{a = x_0 < x_1 < \cdots < x_{n-1} < x_n = b\}$ be a partition of $[a, b]$.

For each $1 \leq i \leq n$, let $t_i \in [x_{i-1}, x_i]$ and call t_i a **sample point**.

The number $S(f, g, P) = \sum_{i=1}^n f(t_i) (g(x_i) - g(x_{i-1}))$ is called a **Riemann-Stieltjes sum of f with respect to g at P** .

We say that f is (**Pollard-**)**Riemann-Stieltjes integrable with respect to g on $[a, b]$**

if there exists a real number A such that for any $\epsilon > 0$,

there is a partition P_ϵ of $[a, b]$ such that for any partition P of $[a, b]$ finer than P_ϵ and any choice of sample points $t_i \in [x_{i-1}, x_i]$,

we have $|S(f, g, P) - A| < \epsilon$,

(which is necessary for that

there is $\delta > 0$ such that for any partition P with $\|P\| < \delta$ and any choice of sample points $t_i \in [x_{i-1}, x_i]$,

we have $|S(f, g, P) - A| < \epsilon$.

Therefore, the standard Riemann integral is a special case of the Riemann-Stieltjes integral where $g(x) = x$.)

In this case, such a number A is unique and is denoted by $\int_a^b f(x)dg(x)$,

where f is called the **integrand**, g is called the **integrator**, and x is called a **dummy variable**.

Theorem 46.2. (*equivalent condition for Riemann-Stieltjes integrability*)

Let $f, g : [a, b] \rightarrow \mathbb{R}$ be bounded.

Then f is Riemann-Stieltjes integrable with respect to g on $[a, b]$

$\iff$ for any $\epsilon > 0$, there exists a partition P of $[a, b]$ such that $U(f, g, P) - L(f, g, P) < \epsilon$,

where $U(f, g, P) = \sum_{i=1}^n \sup\{f(x) : x_{i-1} \leq x \leq x_i\} (g(x_i) - g(x_{i-1}))$

and $L(f, g, P) = \sum_{i=1}^n \inf\{f(x) : x_{i-1} \leq x \leq x_i\} (g(x_i) - g(x_{i-1}))$.

Theorem 46.3. (*Riemann-Stieltjes integrability for step integrator*)

Let $f, g : [a, b] \rightarrow \mathbb{R}$ be bounded functions such that g is a **step function**,

that is, there is a partition $\{a = x_0 < x_1 < \cdots < x_{n-1} < x_n = b\}$ such that g is constant on each open subinterval (x_{i-1}, x_i) ,

and it is not that both f and g are discontinuous from the right or from the left at each x_i .

Then f is Riemann-Stieltjes integrable with respect to g on $[a, b]$, and $\int_a^b f(x)dg(x) = \sum_{i=0}^{n-1} f(x_i)(g(x_{i+1}) - g(x_i))$,

where $g(x_0-) = g(x_0)$ and $g(x_n+) = g(x_n)$.

Theorem 46.4. (*Riemann-Stieltjes integrability for bounded variation integrator*)

Let $f, g : [a, b] \rightarrow \mathbb{R}$ be bounded functions such that g is a **bounded variation**,

that is, there exists $M > 0$ such that for any partition $\{a = x_0 < x_1 < \cdots < x_{n-1} < x_n = b\}$,

one has $\sum_{i=1}^n |g(x_i) - g(x_{i-1})| \leq M$,

and f is continuous on $[a, b]$.

Then f is Riemann-Stieltjes integrable with respect to g on $[a, b]$

and g is Riemann-Stieltjes integrable with respect to f on $[a, b]$.

Theorem 46.5.

Let $f, g : [a, b] \rightarrow \mathbb{R}$ be bounded such that

f is Riemann-Stieltjes integrable with respect to g on $[a, b]$, and g is continuously differentiable on $[a, b]$

Then $\int_a^b f(x)dg(x) = \int_a^b f(x)g'(x)dx$.